

Math 220B HW 2

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1 D

Problem 1

Let R be a semilocal ring and let M and N be two R -modules such that $M \otimes_R N = 0$. Prove or disprove that either $M = 0$ or $N = 0$.

Solution: We'll disprove the statement by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, which it is semilocal ring because it only has finitely many maximal ideals (in fact, finitely many ideals since the ring is finite).

Now, consider the canonical projection $\pi_2 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\pi_3 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$, notice that since $6\mathbb{Z} = 2\mathbb{Z} \cap 3\mathbb{Z}$, one has the ideal $6\mathbb{Z} \subset \ker(\pi_2) = 2\mathbb{Z}$ and $6\mathbb{Z} \subset \ker(\pi_3) = 3\mathbb{Z}$. Hence, by Generalized First Isomorphism Theorem, with the canonical projection $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, the above two maps uniquely factor into surjective maps $\varphi_2 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\varphi_3 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ as follow:

$$\begin{array}{ccc}
 \mathbb{Z} & \xrightarrow{\pi_2} & \mathbb{Z}/2\mathbb{Z} \\
 \searrow \pi_6 & & \nearrow \varphi_2 \\
 & \mathbb{Z}/6\mathbb{Z} &
 \end{array}
 \qquad
 \begin{array}{ccc}
 \mathbb{Z} & \xrightarrow{\pi_3} & \mathbb{Z}/3\mathbb{Z} \\
 \searrow \pi_6 & & \nearrow \varphi_3 \\
 & \mathbb{Z}/6\mathbb{Z} &
 \end{array}$$

Hence, $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$ can be realized as R -module, via the map φ_2 and φ_3 (in particular, one has $\varphi_2(\bar{1}_6) = \bar{1}_2$ and $\varphi_3(\bar{1}_6) = \bar{1}_3$ as the map, based on the above commutative diagrams).

Now, consider the tensor $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z}$, we claim that it is 0: Recall that $2, 3 \in \mathbb{Z}$ are coprime, with Bezout's Lemma there exists integers $s, t \in \mathbb{Z}$, such that $1 = 2s + 3t$. Then, under the projection map $\pi_6 : \mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$, one knows there exists $\bar{s}_6, \bar{t}_6 \in \mathbb{Z}/6\mathbb{Z}$, such that the following holds:

$$\bar{1}_6 = \bar{2}_6 \cdot \bar{s}_6 + \bar{3}_6 \cdot \bar{t}_6 \quad (1.1)$$

Then, for any $\bar{a}_2 \in \mathbb{Z}/2\mathbb{Z}$ and $\bar{b}_3 \in \mathbb{Z}/3\mathbb{Z}$, one has the following:

$$\begin{aligned} \bar{a}_2 \otimes \bar{b}_3 &= \bar{1}_6 \cdot (\bar{a}_2 \otimes \bar{b}_3) = (\bar{2}_6 \cdot \bar{s}_6 + \bar{3}_6 \cdot \bar{t}_6) \cdot (\bar{a}_2 \otimes \bar{b}_3) \\ &= \bar{s}_6 \cdot ((\bar{2}_6 \cdot \bar{a}_2) \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes (\bar{3}_6 \cdot \bar{b}_3)) \\ &= \bar{s}_6 \cdot ((\bar{0}_2 \bar{a}_2) \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes (\bar{0}_3 \bar{b}_3)) \\ &= \bar{s}_6 \cdot (\bar{0}_2 \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes \bar{0}_3) \\ &= 0 \end{aligned} \quad (1.2)$$

(Note: By definition, one has $\bar{2}_6 \cdot \bar{a}_2 := \varphi_2(\bar{2}_6)\bar{a}_2 = \bar{0}_2 \bar{a}_2 = \bar{0}_2$, and same idea for $\mathbb{Z}/3\mathbb{Z}$).

Hence, this proves that $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z} = 0$. Yet, each $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z} \neq 0$ as R -module, which is a desired counterexample for the problem.

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Problem 2

Let $0 \rightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \rightarrow 0$ be an exact sequence of R -modules. Assume that M' and M'' are finitely generated R -modules. Prove that M is a finitely generated R -modules. Explain if the converse is true.

Solution: We'll first prove the given statement, then prove the converse is false in general, by providing a counterexample.

I. Proof of Statement:

Given the given sequence is exact, then it implies $M' \xrightarrow{u} M$ is injective, hence $M' \cong \text{im}(u) \subset M$, WLOG we'll denote $\text{im}(u)$ as M' , and $M' \xrightarrow{u} M$ as an inclusion map, and recognize all elements in M' as in M also.

Since M', M'' are finitely-generated, there exists $x_1, \dots, x_n \in M'$ and $y_1, \dots, y_m \in M''$, such that $M' = \sum_{i=1}^n Rx_i$ and $M'' = \sum_{j=1}^m Ry_j$. Then, choose $z_1, \dots, z_m \in M$, such that for each index $j \in \{1, \dots, m\}$, one has $v(z_j) = y_j$. We claim that the collection $\{x_1, \dots, x_n, z_1, \dots, z_m\}$ generates M .

For any $x \in M$, since $v(x) \in M''$, there exists $b_1, \dots, b_m \in R$, such that the following holds:

$$v(x) = \sum_{j=1}^m b_j y_j = \sum_{j=1}^m b_j v(z_j) = v\left(\sum_{j=1}^m b_j z_j\right) \quad (2.1)$$

As a result, one has $v\left(x - \sum_{j=1}^m b_j z_j\right) = 0$, or $x - \sum_{j=1}^m b_j z_j \in \ker(v)$.

Now, using exactness of the sequence, one gets $\ker(v) = \text{im}(u) = M'$, hence there exists $a_1, \dots, a_n \in R$, such that $x - \sum_{j=1}^m b_j z_j = \sum_{i=1}^n a_i x_i$ by the finitely generated property of M' .

As a result, we have $x = \sum_{i=1}^n a_i x_i + \sum_{j=1}^m b_j z_j$, hence all element $x \in M$ can be expressed with some R -linear combination of $\{x_1, \dots, x_n, z_1, \dots, z_m\}$, showing M is finitely generated.

II. Counterexample of the Converse:

Let k be a field, consider the ring $R = k[x_1, x_2, \dots]$, which R is a polynomial ring over k , with (countably) infinite indeterminates. First, let's note that R consists of polynomials that are finite sums of monomials, and each monomial can involve only finite indeterminates.

Consider the ideal $I := (x_1, x_2, \dots)$ (ideal generated by all indeterminates). Then, $I, R, R/I$ are all R -modules, in particular R is a finitely generated R -module (since every $r \in R$ can be written as $r \cdot 1$, so R is generated by 1 as an R -module). Moreover, we naturally have the following exact sequence of R -linear maps:

$$0 \rightarrow I \hookrightarrow R \twoheadrightarrow R/I \rightarrow 0$$

And it's simply because the image of the inclusion $I \hookrightarrow R$ is I , and the kernel of projection $R \twoheadrightarrow R/I$ is also I .

Now, we have a short exact sequence, such that the middle R -module is finitely generated. Yet, we claim that I is not a finitely generated R -module:

Suppose the contrary that I is a finitely generated R -module, then there exists $f_1, \dots, f_n \in I$, such that $I = \sum_{i=1}^n Rf_i = (f_1, \dots, f_n)$ as ideal. Based on our definition, there are only finitely many indeterminates involved in $f_1, \dots, f_n$, say $S = \{x_{i_1}, \dots, x_{i_k}\}$ (and say it's ordered such that $i_1 < \dots < i_k$).

Then, consider the element $x_{i_{k+1}} \in I$: By our assumption, there exists polynomials $g_1, \dots, g_n \in R$, such that $x_{i_{k+1}} = \sum_{i=1}^n g_i f_i$. Which, let $T = \{x_{j_1}, \dots, x_{j_l}\}$ be the (finite) indeterminates involved in the polynomials $g_1, \dots, g_n$. Which, if consider the set of indeterminates $S \cup T \cup \{x_{i_{k+1}}\} = \{x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}\}$ after reordering, one can realize the elements $x_{i_{k+1}}, f_1, \dots, f_n, g_1, \dots, g_n$ as elements in $k[x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}] \hookrightarrow k[x_1, x_2, \dots]$, and inside this smaller subring, the equality $x_{i_{k+1}} = \sum_{i=1}^n g_i f_i$ still holds.

Now, consider the evaluation map $k[x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}] \rightarrow k$, by $x_{i_{k+1}} \mapsto 1$, and all $x_{m_r} \mapsto 0$: Since we've chosen $x_{i_{k+1}}$, such that it's distinct from all x_{i_j} ($1 \leq j \leq k$), then one has x_{i_j} being some of the x_{m_r} , hence $x_{i_j} \mapsto 0$. Also, notice that since each $f_i \in I$, then it has no constant term (since one has $f_i = h_1 x_1 + \dots$ for $h_i \in R$, such that finitely many $h_i \neq 0$; then, all the constant terms in h_i are multiplied with x_i , which is no longer with constant term). Hence, with each $x_{i_j} \mapsto 0$, $f_i \mapsto 0$ also (since it's a polynomial involving only $x_{i_1}, \dots, x_{i_k}$, and with no constant term).

As a result, one has the following:

$$x_{i_{k+1}} = \sum_{i=1}^n g_i f_i \mapsto 0 \tag{2.2}$$

because each $f_i \mapsto 0$. However, this contradicts our assumption that $x_{i_{k+1}} \mapsto 1$. Hence, our assumption must be false, the ideal I cannot be a finitely generated R -module.

Hence, we found an exact sequence of R -modules $0 \rightarrow I \hookrightarrow R \twoheadrightarrow R/I \rightarrow 0$, such that the middle is finitely generated R -module, while the two sides are not guaranteed to be finitely generated (in particular, the left side is not guaranteed). This proves that the converse is in general false.

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Problem 3

Let M and N be flat R -modules, where R is a commutative ring. Prove or disprove that $M \otimes_R N$ is flat R -module.

Solution: We'll prove that $M \otimes_R N$ is also a flat R -module.

Since the tensor functor $(_) \otimes_R (M \otimes_R N) : R\text{-Mod} \rightarrow R\text{-Mod}$ is a right exact functor, to show that $M \otimes_R N$ is flat (or the above tensor functor is exact), one only needs to prove it preserves injective R -linear maps.

Recall that for any R -module K, L, P , there is a unique R -linear isomorphism $\varphi_{KLP} : (K \otimes_R L) \otimes_R P \xrightarrow{\sim} K \otimes_R (L \otimes_R P)$ that satisfies $\varphi_{KLP}((k \otimes l) \otimes p) = k \otimes (l \otimes p)$ for all $k \in K, l \in L$, and $p \in P$.

Now, given an injective R -linear map $f : K \hookrightarrow K'$, since M is flat, then the exactness of the functor $(_) \otimes_R M$ guarantees the map $f \otimes \text{id}_M : K \otimes_R M \rightarrow K' \otimes_R M$ is injective (which the map is given by $(f \otimes \text{id}_M)(k \otimes m) = f(k) \otimes m$ for all $k \in K, m \in M$).

Similarly, since N is flat, then the exactness of the functor $(_) \otimes_R N$ guarantees the map $(f \otimes \text{id}_M) \otimes \text{id}_N : (K \otimes_R M) \otimes_R N \rightarrow (K' \otimes_R M) \otimes_R N$ is injective (which the map is given by $((f \otimes \text{id}_M) \otimes \text{id}_N)((k \otimes m) \otimes n) = (f(k) \otimes m) \otimes n$ for all $k \in K, m \in M$, and $n \in N$).

Finally, consider the following diagram:

$$\begin{array}{ccc} K \otimes_R (M \otimes_R N) & \xrightarrow{f \otimes \text{id}_{M \otimes_R N}} & K' \otimes_R (M \otimes_R N) \\ \varphi_{KMN}^{-1} \downarrow & & \uparrow \varphi_{K'MN} \\ (K \otimes_R M) \otimes_R N & \xrightarrow{(f \otimes \text{id}_M) \otimes \text{id}_N} & (K' \otimes_R M) \otimes_R N \end{array}$$

Notice that the diagram commutes, because for each $k \in K, m \in M$, and $n \in N$, one has the following:

$$(f \otimes \text{id}_{M \otimes_R N})(k \otimes (m \otimes n)) = f(k) \otimes (m \otimes n) \quad (3.1)$$

$$\begin{aligned} & \varphi_{K'MN} \circ ((f \otimes \text{id}_M) \otimes \text{id}_N) \circ \varphi_{KMN}^{-1}(k \otimes (m \otimes n)) \\ &= \varphi_{K'MN} \circ ((f \otimes \text{id}_M) \otimes \text{id}_N)((k \otimes m) \otimes n) \\ &= \varphi_{K'MN}((f(k) \otimes m) \otimes n) \\ &= f(k) \otimes (m \otimes n) \end{aligned} \quad (3.2)$$

Then, since $K \otimes_R (M \otimes_R N)$ is generated by all elements of the form $k \otimes (m \otimes n)$, then the above two calculation shows that the two maps are the same, hence the diagram commutes.

As a result, since $\varphi_{K'MN}, (f \otimes \text{id}_M) \otimes \text{id}_N$, and φ_{KMN}^{-1} are all injective, then their composition – which results in $f \otimes \text{id}_{M \otimes_R N}$ – is injective. Therefore, the functor $(_) \otimes_R (M \otimes_R N)$ preserves injective maps, hence is exact. This shows that $M \otimes_R N$ is a flat module.

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Problem 4

Let $\varphi : R \rightarrow S$ be a ring homomorphism and let M be a flat R -module. Prove or disprove that $M \otimes_R S$ is a flat S -module.

Solution: We'll prove that given M is a flat R -module, then $M \otimes_R S$ is a flat S -module.

Since $(M \otimes_R S) \otimes_S (-) : \mathbf{S}\text{-Mod} \rightarrow \mathbf{S}\text{-Mod}$ is right exact, one just needs to prove it preserves injective maps.

For this, let's prove an exercise in Atiyah Macdonald Chapter 2:

Exercise: 2.15

Let A, B be commutative rings, let M be an A -module, P a B -module and N an (A, B) -bimodule (that is, N is simultaneously an A -module and B -module and the two structures are compatible in the sense that $a(xb) = (ax)b$ for all $a \in A, b \in B, x \in N$).

Then, $M \otimes_A N$ is naturally a B -module, $N \otimes_B P$ an A -module, and we have:

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P) \quad (4.1)$$

Proof: We'll prove the three claims in sequence:

I. B -Module Structure on $M \otimes_A N$:

For any $b \in B, m \in M$ and $n \in N$, define the action $B \times (M \otimes_A N) \rightarrow M \otimes_A N$ as $b \cdot (m \otimes_A n) := m \otimes_A (nb)$ on each generator $m \otimes_A n$ (where nb is given by the B -module structure of N), we also define its action on any sum $\sum_{i=1}^k m_i \otimes_A n_i$ by sum of its action on individual $m_i \otimes_A n_i$. As a set map this is well-defined, let's check it does satisfy the B -module properties (which suffices to check for the generators of $M \otimes_A N$). Given any $b, b' \in B$, and any element $m \otimes_A n, m' \otimes_A n' \in M \otimes_A N$, we have the following:

$$\begin{aligned} 1. \quad (b + b') \cdot (m \otimes_A n) &= m \otimes_A (n(b + b')) = m \otimes_A (nb + nb') \\ &= m \otimes_A (nb) + m \otimes_A (nb') \end{aligned} \quad (4.2)$$

$$\begin{aligned} 2. \quad b \cdot (m \otimes_A n + m' \otimes_A n') &= b \cdot (m \otimes_A n) + b \cdot (m' \otimes_A n') \end{aligned} \quad (4.3)$$

(Note: This is by our definition, such that the action on the sum, is the sum of each individual generator after the action).

$$\begin{aligned} 3. \quad (bb') \cdot (m \otimes_A n) &= m \otimes_A (n(bb')) = m \otimes_A (n(b'b)) \\ &= m \otimes_A ((nb')b) = b \cdot (m \otimes_A (nb')) \\ &= b \cdot (b' \cdot (m \otimes_A n)) \end{aligned} \quad (4.4)$$

$$4. \quad 1_B \cdot (m \otimes_A n) = m \otimes_A (n1_B) = m \otimes_A n \quad (4.5)$$

These are all the requirements for being a B -module, hence $M \otimes_A N$ has a B -module structure (by its action on N).

As a side note, this in fact makes $M \otimes_A N$ an (A, B) -bimodule structure, since for any $m \otimes_A n \in M \otimes_A N, a \in A$, and $b \in B$, we have the following:

$$\begin{aligned} a \cdot (b \cdot (m \otimes_A n)) &= a \cdot (m \otimes_A (nb)) = m \otimes_A (a(nb)) \\ &= m \otimes_A ((an)b) = b \cdot (m \otimes_A (an)) \\ &= b \cdot (a \cdot (m \otimes_A n)) \end{aligned} \quad (4.6)$$

Which, the A -action and B -action commutes, showing the bimodule property.

II. A -Module Structure on $N \otimes_B P$:

Similarly, define $A \times (N \otimes_B P) \rightarrow N \otimes_B P$ by $a \cdot (n \otimes_B p) := (an) \otimes_B p$. Based on similar reasonings in **I**, it satisfies all the conditions for A -module, hence $N \otimes_B P$ now can be realized as A -module.

Again, similar to the proof in **I**, the computation shows that $N \otimes_B P$ is an (A, B) -bimodule.

III. Isomorphism:

We aim to show that $(M \otimes_A N) \otimes_B P$ and $M \otimes_A (N \otimes_B P)$ are isomorphic as both A and B -module. For this, we'll construct a B -linear map $(M \otimes_A N) \otimes_B P \rightarrow M \otimes_A (N \otimes_B P)$, and another A -linear map $(M \otimes_A N) \otimes_B P \leftarrow M \otimes_A (N \otimes_B P)$, and prove that these two maps are isomorphism for abelian groups, while both maps are A -linear and B -linear, which will complete the isomorphism.

1. B -linear map:

First, fix arbitrary $p \in P$, and define a map $F_p : M \times N \rightarrow M \otimes_A (N \otimes_B P)$ by $(m, n) \mapsto m \otimes_A (n \otimes_B p)$ for all $(m, n) \in M \times N$. Note that this is A -bilinear, since for all $a, a' \in A$, $m, m' \in M$, and $n, n' \in N$, we have the following:

$$\begin{aligned} F_p(am + a'm', n) &= (am + a'm') \otimes_A (n \otimes_B p) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m' \otimes_A (n \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m', n) \end{aligned} \quad (4.7)$$

$$\begin{aligned} F_p(m, an + a'n') &= m \otimes_A ((an + a'n') \otimes_B p) \\ &= m \otimes_A (a \cdot (n \otimes_B p) + a' \cdot (n' \otimes_B p)) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m \otimes_A (n' \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m, n') \end{aligned} \quad (4.8)$$

Hence, this uniquely factors through the tensor product, as $\overline{F}_p : M \otimes_A N \rightarrow M \otimes_A (N \otimes_B P)$ by $\overline{F}_p(m \otimes_A n) = m \otimes_A (n \otimes_B p)$.

Now, let's define a map $G : (M \otimes_A N) \times P \rightarrow M \otimes_A (N \otimes_B P)$ as this:

$$G\left(\sum_{i=1}^k m_i \otimes_A n_i, p\right) := \overline{F}_p\left(\sum_{i=1}^k m_i \otimes_A n_i\right) \quad (4.9)$$

We claim that G is B -bilinear (which suffices to check each individual tensors). Given any $b, b' \in B$, $(m \otimes_A n), (m' \otimes_A n') \in M \otimes_A N$, and $p, p' \in P$, we have the following:

$$\begin{aligned}
G(b \cdot (m \otimes_A n) + b' \cdot (m' \otimes_A n'), p) &= \overline{F}_p(m \otimes_A (nb) + m' \otimes_A (n'b')) \\
&= m \otimes_A ((nb) \otimes_B p) + m' \otimes_A ((n'b') \otimes_B p) \\
&= m \otimes_A ((n \otimes_B p)b) + m' \otimes_A ((n' \otimes_B p)b') \\
&= b \cdot (m \otimes_A (n \otimes_B p)) + b' \cdot (m' \otimes_A (n' \otimes_B p)) \quad (4.10) \\
&= b \cdot \overline{F}_p(m \otimes_A n) + b' \cdot \overline{F}_p(m' \otimes_A n') \\
&= b \cdot G(m \otimes_A n, p) + b' \cdot G(m' \otimes_A n', p)
\end{aligned}$$

$$\begin{aligned}
G(m \otimes_A n, bp + b'p') &= F_{bp+b'p'}(m \otimes_A n) \\
&= m \otimes_A (n \otimes_B (bp + b'p')) \\
&= m \otimes_A (b \cdot (n \otimes_B p) + b' \cdot (n \otimes_B p')) \\
&= b \cdot (m \otimes_A (n \otimes_B p)) + b' \cdot (m \otimes_A (n \otimes_B p')) \quad (4.11) \\
&= b \cdot \overline{F}_p(m \otimes_A n) + b' \cdot \overline{F}_{p'}(m \otimes_A n) \\
&= b \cdot G(m \otimes_A n, p) + b' \cdot G(m \otimes_A n, p')
\end{aligned}$$

Since G is B -bilinear, there exists a unique B -linear map $\overline{G} : (M \otimes_A N) \otimes_B P \rightarrow M \otimes_A (N \otimes_B P)$ by $\overline{G}((m \otimes_A n) \otimes_B p) = m \otimes_A (n \otimes_B p)$.

2. A -linear map:

Similarly, fix arbitrary $m \in M$, and define a map $H_m : N \times P \rightarrow (M \otimes_A N) \otimes_B P$ by $(n, p) \mapsto (m \otimes_A n) \otimes_B p$ for all $(n, p) \in N \times P$. Using the same logic in the previous part, one can prove that H_m is B -bilinear, hence inducing a unique B -linear map $\overline{H}_m : N \otimes_B P \rightarrow (M \otimes_A N) \otimes_B P$ by $\overline{H}_m(n \otimes_B p) = m \otimes_A (n \otimes_B p)$.

Then, one defines another map $L : M \times (N \otimes_B P) \rightarrow (M \otimes_A N) \otimes_B P$ as follow:

$$L\left(m, \sum_{i=1}^k n_i \otimes_B p_i\right) := \overline{H}_m\left(\sum_{i=1}^k n_i \otimes_B p_i\right) \quad (4.12)$$

Similar to the previous part, we in fact have L being A -bilinear after running through the computation. Hence, this induces a unique A -linear map $\overline{L} : M \otimes_A (N \otimes_B P) \rightarrow (M \otimes_A N) \otimes_B P$ by $\overline{L}(m \otimes_A (n \otimes_B p)) = (m \otimes_A n) \otimes_B p$.

3. Isomorphism:

Given the B -linear map $\overline{G} : (M \otimes_A N) \otimes_B P \rightarrow M \otimes_A (N \otimes_B P)$, and the A -linear map $\overline{L} : M \otimes_A (N \otimes_B P) \rightarrow (M \otimes_A N) \otimes_B P$, if we first look at the two modules' abelian group structure, one have this for all $m \in M, n \in N, p \in P$:

$$\begin{aligned}
\overline{L} \circ \overline{G}((m \otimes_A n) \otimes_B p) &= \overline{L}(m \otimes_A (n \otimes_B p)) = (m \otimes_A n) \otimes_B p \\
\overline{G} \circ \overline{L}(m \otimes_A (n \otimes_B p)) &= \overline{G}((m \otimes_A n) \otimes_B p) = m \otimes_A (n \otimes_B p)
\end{aligned} \quad (4.13)$$

Hence, as homomorphism of abelian groups, $\overline{G}, \overline{L}$ are mutual inverses (since the compositions act as identity on all the generators of the two spaces, in the right composition order).

Now, it suffices to check that $\overline{G}, \overline{L}$ are in fact both A -linear and B -linear.

- For $\overline{G}$, we've constructed it as a B -linear map, so it suffices to prove the A -linear structure (in particular, suffices to prove it preserves the A -action, since additivity is given by the fact that $\overline{G}$ is a homomorphism of abelian group). Given any $a \in A$, and $(m \otimes_A n) \otimes_B p \in (M \otimes_A N) \otimes_B P$, one has the following:

$$\begin{aligned}
\overline{G}(a \cdot ((m \otimes_A n) \otimes_B p)) &= \overline{G}((a \cdot (m \otimes_A n)) \otimes_B p) \\
&= \overline{G}((am) \otimes_A n \otimes_B p) \\
&= (am) \otimes_A (n \otimes_B p) \\
&= a \cdot (m \otimes_A (n \otimes_B p)) \\
&= a \cdot \overline{G}((m \otimes_A n) \otimes_B p)
\end{aligned} \tag{4.14}$$

(Note: Here we use the fact that $(M \otimes_A N) \otimes_B P$ has an A -module structure by letting A act on $M \otimes_A N$, which is proven in **I**, because M, N can be arbitrary).

Hence, $\overline{G}$ preserves A -action, which is A -linear.

- Similarly, for $\overline{L}$, it's constructed as an A -linear map, so it suffices to prove it preserves the B -action. Given any $b \in B$, and $m \otimes_A (n \otimes_B p) \in M \otimes_A (N \otimes_B P)$, one has the following:

$$\begin{aligned}
\overline{L}(b \cdot (m \otimes_A (n \otimes_B p))) &= \overline{L}(m \otimes_A (b \cdot (n \otimes_B p))) \\
&= \overline{L}(m \otimes_A (n \otimes_B (bp))) \\
&= (m \otimes_A n) \otimes_B (bp) \\
&= b \cdot ((m \otimes_A n) \otimes_B p) \\
&= b \cdot \overline{L}(m \otimes_A (n \otimes_B p))
\end{aligned} \tag{4.15}$$

(Note: Again, we use the fact that $M \otimes_A (N \otimes_B P)$ has B -module structure by letting B act on $N \otimes_B P$, proven in **II**).

Hence, $\overline{L}$ preserves B -action, which is a B -linear map.

This realizes $\overline{G}, \overline{L}$ as mutual inverses as both A -linear and B -linear map, hence proving that $(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)$, as (A, B) -bimodule. In particular, an isomorphism (as both A and B -module) between them that's constructed above, has the following form:

$$\begin{aligned}
\varphi_{MNP} : (M \otimes_A N) \otimes_B P &\xrightarrow{\sim} M \otimes_A (N \otimes_B P) \\
\varphi_{MNP}((m \otimes_A n) \otimes_B p) &= m \otimes_A (n \otimes_B p), \quad \forall m \in M, n \in N, p \in P
\end{aligned} \tag{4.16}$$

□

Since S is an (R, S) -bimodule (where it's both an R -module and S -module, while any $r \in R, s, t \in S$ has $(r \cdot t)s = (\varphi(r)t)s = \varphi(r)(ts) = r \cdot (ts)$, where the dot $\cdot$ represents the R -action on S), then one has the following isomorphism, for any S -module P :

$$\varphi_{MSP} : (M \otimes_R S) \otimes_S P \xrightarrow{\sim} M \otimes_R (S \otimes_S P), \quad \varphi_{MSP}((m \otimes_R s) \otimes_S p) = m \otimes_R (s \otimes_S p) \tag{4.17}$$

Now, given any injective S -linear map $f : P \hookrightarrow P'$ between two S -modules, one simply has $(\text{id}_S) \otimes_S f : S \otimes_S P \hookrightarrow S \otimes_S P'$ be injective (since tensor an S -module over S is isomorphic to the original module).

However, $(\text{id}_S) \otimes_S f$ is also an injective R -linear map, since it's additive, and for all $r \in R, s \in S$, and $p \in P$, one has the following:

$$\begin{aligned}
((\text{id}_S) \otimes_S f)(r \cdot (s \otimes_B p)) &= ((\text{id}_S) \otimes_S f)((r \cdot s) \otimes_B p) \\
&= (r \cdot s) \otimes_B f(p) \\
&= r \cdot (s \otimes_B f(p)) \\
&= r \cdot ((\text{id}_S) \otimes_S f)(s \otimes_B p)
\end{aligned} \tag{4.18}$$

Hence, with M being a flat R -module, the R -linear map $\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f) : M \otimes_R (S \otimes_S P) \hookrightarrow M \otimes_R (S \otimes_S P')$ is also injective.

Similar to the previous one, we also have $\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f)$ being an injective S -linear map, since it's additive, for any $m \in M$, $s, t \in S$, and $p \in P$, one has the following:

$$\begin{aligned}
(\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f))(s \cdot (m \otimes_R (t \otimes_S p))) &= (\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f))(m \otimes_R (s \cdot (t \otimes_S p))) \\
&= m \otimes_R (((\text{id}_S) \otimes_S f)((st) \otimes_S p)) \\
&= m \otimes_R ((st) \otimes_S f(p)) \\
&= m \otimes_R (s \cdot (t \otimes_S f(p))) \\
&= s \cdot (m \otimes_R (t \otimes_S f(p))) \\
&= s \cdot [(\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f))(m \otimes_R (t \otimes_S p))]
\end{aligned} \tag{4.19}$$

For simplicity, let's define $\psi := \text{id}_M \otimes_R ((\text{id}_S) \otimes_S f)$ as the injective S -linear map.

Finally, one has the following diagram:

$$\begin{array}{ccc}
(M \otimes_R S) \otimes_S P & \xrightarrow{(\text{id}_M \otimes_R S) \otimes_S f} & (M \otimes_R S) \otimes_S P' \\
\varphi_{MSP} \downarrow & & \uparrow \varphi_{MSP'}^{-1} \\
M \otimes_R (S \otimes_S P) & \xrightarrow{\psi} & M \otimes_R (S \otimes_S P')
\end{array}$$

Notice that this big diagram commutes, because for all $m \in M$, $s \in S$, and $p \in P$, one has the following:

$$((\text{id}_M \otimes_R S) \otimes_S f)((m \otimes_R s) \otimes_S p) = (m \otimes_R s) \otimes_S f(p) \tag{4.20}$$

$$\begin{aligned}
\varphi_{MSP'}^{-1} \circ \psi \circ \varphi_{MSP}((m \otimes_R s) \otimes_S p) &= \varphi_{MSP'}^{-1} \circ \psi(m \otimes_R (s \otimes_S p)) \\
&= \varphi_{MSP'}^{-1}(m \otimes_R (s \otimes_S f(p))) \\
&= (m \otimes_R s) \otimes_S f(p)
\end{aligned} \tag{4.21}$$

This shows that $\varphi_{MSP'}^{-1} \circ \psi \circ \varphi_{MSP} = (\text{id}_M \otimes_R S) \otimes_S f$ as S -linear maps. Hence, with each map on the LRS being injective, the RHS is also injective.

This shows that $(M \otimes_R S) \otimes_S (_)$ is an exact functor (as it preserves injectivity), hence $M \otimes_R S$ can be realized as a flat S -module.

5 D

Problem 5

Let $\varphi : R \rightarrow S$ be a ring homomorphism such that M is projective as an R -module. Prove or disprove that $M \otimes_R S$ is projective as S -module.

Solution: We'll prove that if M is projective as R -module, then $M \otimes_R S$ is projective as S -module.

Since M is projective as an R -module, there exists another R -module N , such that $M \oplus N \cong \bigoplus_{i \in I} Re_i$ a free R -module (where each $Re_i \cong R$ as R -module). Then, since tensor product „commutes“ with direct sum (up to isomorphism), hence one has the following isomorphism as R -modules:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong (M \oplus N) \otimes_R S \cong \left(\bigoplus_{i \in I} Re_i \right) \otimes_R S \cong \bigoplus_{i \in I} (Re_i \otimes_R S) \quad (5.1)$$

As a remark, all these isomorphism respects the S -module structure: Given any family of R -modules $\{M_i\}_{i \in \Lambda}$, the R -linear isomorphism $\varphi : \left(\bigoplus_{i \in \Lambda} M_i \right) \otimes_R S \xrightarrow{\sim} \bigoplus_{i \in \Lambda} (M_i \otimes_R S)$ is given as:

$$\varphi\left(\left((m_i)_{i \in \Lambda}\right) \otimes s\right) = (m_i \otimes s)_{i \in \Lambda} \quad (5.2)$$

Which, this map is also S -linear, as any $(m_i)_{i \in \Lambda} \in \bigoplus_{i \in \Lambda} M_i$, and $s, t \in S$, satisfy the following:

$$\begin{aligned} \varphi\left(s \cdot \left((m_i)_{i \in \Lambda} \otimes t\right)\right) &= \varphi\left((m_i)_{i \in \Lambda} \otimes (ts)\right) = (m_i \otimes (ts))_{i \in \Lambda} \\ &= (s \cdot (m_i \otimes t))_{i \in \Lambda} = s \cdot (m_i \otimes t)_{i \in \Lambda} \end{aligned} \quad (5.3)$$

Hence, φ is also an S -linear isomorphism. Apply this logic to the 1st and 3rd isomorphism relation, they're isomorphic as S -module (the middle part is true because S is preserved).

Now, look at each $Re_i \cong R$, one has each $Re_i \otimes_R S \cong S$ as R -module (and a specific R -linear isomorphism is given by $\varphi : R \otimes_R S \xrightarrow{\sim} S$, $\varphi(r \otimes s) = r \cdot s = \varphi(r)s$).

However, notice that this φ is also an S -linear map, because it's additive, and also every $r \in R$ and $s, t \in S$ satisfy the following:

$$\varphi(s \cdot (r \otimes t)) = \varphi(r \otimes (ts)) = r \cdot (ts) = \varphi(r)(ts) = (\varphi(r)t)s = s \cdot (r \cdot t) = s \cdot \varphi(r \otimes t) \quad (5.4)$$

(Note: Here the S -module structure on $R \otimes_R S$ is by letting S act on S).

Hence, it's not only an R -linear isomorphism, but in fact an S -linear isomorphism. As a consequence, with the S -module structure on $R \otimes_R S$ being isomorphic to S , one has:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong \bigoplus_{i \in I} (Re_i \otimes_R S) \cong \bigoplus_{i \in I} S \quad (5.5)$$

Where the final part is a free S -module. This realizes $M \otimes_R S$ as a direct summand of a free S -module, hence it's projective as S -module.

6 HD (Type it up)

Problem 6

Let $\varphi : R \rightarrow S$ be a ring homomorphism and let M be an injective R -module such that $M \otimes_R S \neq 0$. Prove or disprove that $M \otimes_R S$ is injective S -module.

Solution: Counterexample: $R = \mathbb{Z}$, $S = \mathbb{Z}[x]$, and $M = \mathbb{Q}$ injective $\mathbb{Z}$ -mod. Then, prove that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}[x] \cong \mathbb{Q}[x]$ as $\mathbb{Z}$ -algebra, and hence the $\mathbb{Z}[x]$ -module structure is given by $\mathbb{Z}[x] \hookrightarrow \mathbb{Q}[x]$. But, $\mathbb{Q}[x]$ is not injective $\mathbb{Z}[x]$ -module, because it's not divisible.

7 D

Problem 7

Let $\varphi : R \rightarrow S$ be a ring homomorphism such that S is flat R -algebra. Prove or disprove that φ is injective.

Solution: We'll disprove it by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, recall that it has a projection $\varphi : R \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ by $\varphi(\bar{1}_6) = \bar{1}_3$, this realizes $\mathbb{Z}/3\mathbb{Z}$ as an R -algebra, while the map φ is not injective (since $\varphi(\bar{3}_6) = \bar{3}_3 = 0$).

Now, consider the ideal $I = \{\bar{0}_6, \bar{2}_6, \bar{4}_6\} \subset R$, which it can be obtained using another projection $\pi_2 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ by $\pi_2(\bar{n}_6) = \bar{n}_2$. Which, $\bar{n}_2 = \bar{0}_2$ iff n is odd iff $\bar{n}_6 \in \{\bar{0}_6, \bar{2}_6, \bar{4}_6\}$, showing $\ker(\varphi) = I$.

However, this ideal also has a ring structure: It's closed under multiplication by the ideal property, so it suffices to find the inverse. Consider the following multiplication relation:

$$\bar{0}_6 \cdot \bar{4}_6 = \bar{0}_6, \quad \bar{2}_6 \cdot \bar{4}_6 = \bar{8}_6 = \bar{2}_6, \quad \bar{4}_6 \cdot \bar{4}_6 = \bar{16}_6 = \bar{4}_6 \quad (7.1)$$

This shows that all $\bar{a}_6 \in I$ satisfies $\bar{a}_6 \cdot \bar{4}_6 = \bar{a}_6$, which $\bar{4}_6$ is the multiplicative identity of I .

Notice that I as a ring with the above structure is isomorphic to $\mathbb{Z}/3\mathbb{Z}$: As an additive abelian group because both I and $\mathbb{Z}/3\mathbb{Z}$ has order 3, they must be isomorphic, and one abelian group isomorphism can be considered as $j : \mathbb{Z}/3\mathbb{Z} \xrightarrow{\sim} I$, $j(\bar{1}_3) = \bar{4}_6$ (since both are cyclic with prime order, then any nonzero element is a generator).

Yet, j is also a ring homomorphism, because of the following (for anything multiplied by zero, it's sent to zero by group homomorphism property, so we don't need to check):

$$j(\bar{1}_3 \cdot \bar{2}_3) = j(\bar{2}_3) = \bar{2}_6 = \bar{4}_6 \cdot \bar{2}_6 = j(\bar{1}_3) \cdot j(\bar{2}_3) \quad (7.2)$$

$$j(\bar{1}_3 \cdot \bar{1}_3) = j(\bar{1}_3) = \bar{4}_6 = \bar{4}_6 \cdot \bar{4}_6 = j(\bar{1}_3) \cdot j(\bar{1}_3) \quad (7.3)$$

$$j(\bar{2}_3 \cdot \bar{2}_3) = j(\bar{1}_3) = \bar{4}_6 = \bar{2}_6 \cdot \bar{2}_6 = j(\bar{2}_3) \cdot j(\bar{2}_3) \quad (7.4)$$

Hence, this realizes $\mathbb{Z}/3\mathbb{Z} \cong I$ as rings, and even stronger – as R -algebras. This suffices to check the R -action on $\bar{1}_3$ (as the rest follows directly by adding $\bar{1}_3$ to itself):

$$\begin{aligned} j(\bar{1}_6 \cdot \bar{1}_3) &= j(\bar{1}_3) = \bar{4}_6 = \bar{1}_6 \cdot \bar{4}_6 = \bar{1}_6 \cdot j(\bar{1}_3) \\ j(\bar{2}_6 \cdot \bar{1}_3) &= j(\bar{2}_3) = \bar{2}_6 = \bar{2}_6 \cdot \bar{4}_6 = \bar{2}_6 \cdot j(\bar{1}_3) \\ j(\bar{3}_6 \cdot \bar{1}_3) &= j(\bar{0}_3) = \bar{0}_6 = \bar{3}_6 \cdot \bar{4}_6 = \bar{3}_6 \cdot j(\bar{1}_3) \\ j(\bar{4}_6 \cdot \bar{1}_3) &= j(\bar{1}_3) = \bar{4}_6 = \bar{4}_6 \cdot \bar{4}_6 = \bar{4}_6 \cdot j(\bar{1}_3) \\ j(\bar{5}_6 \cdot \bar{1}_3) &= j(\bar{2}_3) = \bar{2}_6 = \bar{5}_6 \cdot \bar{4}_6 = \bar{1}_6 \cdot j(\bar{1}_3) \end{aligned} \quad (7.5)$$

This proves the R -linearity of j , hence $j : \mathbb{Z}/3\mathbb{Z} \xrightarrow{\sim} I$ is in fact an R -linear isomorphism.

As a result, this is actually a section of the map $\varphi : R \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ when viewed as an R -linear map:

$$\varphi \circ j(\bar{1}_3) = \varphi(\bar{4}_6) = \bar{1}_3 \quad (7.6)$$

And, with $\bar{1}_3$ being a generator of $\mathbb{Z}/3\mathbb{Z}$ additive wise, this proves that $\varphi \circ j = \text{id}_{\mathbb{Z}/3\mathbb{Z}}$. Hence, the R -linear map $\varphi : R \rightarrow \mathbb{Z}/3\mathbb{Z}$ splits, showing that $\mathbb{Z}/3\mathbb{Z}$ is a direct summand of R (which is a free R -module).

This proves that $\mathbb{Z}/3\mathbb{Z}$ is a projective R -module, which in particular is flat. However, as a flat R -algebra, the map $\varphi : R \rightarrow \mathbb{Z}/3\mathbb{Z}$ is not injective, which provides a counterexample to our statement.

8 D

Problem 8

Let $\varphi : R \rightarrow S$ be an injective ring homomorphism such that S is a field. Prove or disprove that S is a flat R -algebra.

Solution: We'll prove that S is a flat R -algebra.

First, since $\varphi : R \hookrightarrow S$ is injective, we can identify R as a subring of S , and φ as a simple inclusion. As a side note, R is an integral domain.

Now, consider any inclusion of nonzero ideal $\iota : I \hookrightarrow R$, and consider its map after tensoring with S over R , say $\iota \otimes \text{id}_S : I \otimes_R S \rightarrow R \otimes_R S$. The goal is to prove this map is injective (in fact, we'll prove that it's an isomorphism): Consider the isomorphism as R -modules $R \otimes_R S \simeq S$ by $r \otimes s \mapsto r \cdot s$. Then, $\iota \otimes \text{id}_S$ can also be identified as follow:

$$\iota \otimes \text{id}_S : I \otimes_R S \rightarrow S, \quad (\iota \otimes \text{id}_S)(a \otimes s) = as \quad (8.1)$$

Which, fix a nonzero $a \in I$, define the map $g : S \rightarrow I \otimes_R S$ by $g(s) := a \otimes \frac{s}{a}$. Notice that this is well-defined, as for any nonzero $a, b \in I$ and any $s \in S$, one has the following:

$$a \otimes \frac{s}{a} = a \otimes \frac{bs}{ba} = (ab) \otimes \frac{s}{ba} = b \otimes \frac{as}{ba} = b \otimes \frac{s}{b} \quad (8.2)$$

Hence, the image is in fact independent of the choice in I . Now, notice that g is also an R -linear map. For all $r \in R$ and $s, t \in S$, one has the following:

$$g(s+t) = a \otimes \frac{s+t}{a} = a \otimes \left(\frac{s}{a} + \frac{t}{a} \right) = a \otimes \frac{s}{a} + a \otimes \frac{t}{a} = g(s) + g(t) \quad (8.3)$$

$$g(r \cdot s) = a \otimes \frac{rs}{a} = r \cdot \left(a \otimes \frac{s}{a} \right) = r \cdot g(s) \quad (8.4)$$

Finally, recognize that g is a mutual inverse of $\iota \otimes \text{id}_S$, as for all $s \in S$ and $b \in I$, one has the following;

$$g \circ (\iota \otimes \text{id}_S)(b \otimes s) = g(bs) = a \otimes \frac{bs}{a} = (ab) \otimes \frac{s}{a} = b \otimes \frac{as}{a} = b \otimes s \quad (8.5)$$

(Note: since $b \in I$, then with ab on the left one can factor out a , and still have $b \otimes \frac{as}{a} \in I \otimes_R S$).

$$(\iota \otimes \text{id}_S) \circ g(s) = (\iota \otimes \text{id}_S) \left(a \otimes \frac{s}{a} \right) = a \cdot \frac{s}{a} = s \quad (8.6)$$

This shows that $\iota \otimes \text{id}_S, g$ are mutual inverses on the generators of $I \otimes_R S$ and S , hence are mutual inverses on the two R -modules.

This proves injectivity of $\iota \otimes \text{id}_S$. And, since $I \hookrightarrow R$ is arbitrary inclusion of nonzero ideals, preserving injectivity for all such inclusion of nonzero ideals under tensor implies S is a flat R -algebra.

9 HD (Type it up)

Problem 9

Let R be a dedekind domain and let $I \subset R$ be an ideal. Prove or disprove that I is a flat R -module.

Solution: Over a Dedekind Domain, a module is torsion free $\iff$ it's flat (look up how localization makes this work, it's in 3/5's notes). Hence, any $I \subset R$ ideal is torsion free, hence flat.

First, prove that under localization of a dedekind domain, every ideal must be some product of prime ideals (by pulling it back to the dedekind domain itself, and use the unique factorization property). Then, use the fact that in a dedekind domain, all nonzero prime ideal is actually maximal, hence this reduces the ideal to a power of the maximal ideal (in the localization).

Now, if $m^2 = m$ (m the maximal ideal), bc it's actually an R -module (with m being the maximal ideal), then Nakayama's Lemma states that $m^2 = m$ implies $m = 0$, or R is a field (i.e. localize at 0).

Else if $m^2 \subsetneq m$, find $r \in m \setminus m^2$, we claim that $(r) = m$: It's clear $(r) \subseteq m$ by definition; for the reverse,

10 ND (Need other ways to get through the classification theorem)

Problem 10

Let R be a PID and let M be a finitely generated flat R -module. Prove or disprove that M is a free R -module.

Solution: Known information: All submodules of a finitely generated free module over R (a PID) is finitely generated and free.

Which, the quotient is of the form $R^n \oplus \frac{R}{I_1} \oplus \dots \oplus \frac{R}{I_t}$. So, it suffices to prove that if M is of such form, it must be free (or prove that some R/I over a PID can be flat).

However, R/I is not flat (since it's torsion), which implies it can't be in the direct summand, which must be just R^n , which is free.

11 D

Problem 11

Let R be a PID and let M be an R -module. Prove that M is a submodule of an injective R -module.

Solution: Recall that over R a PID, injective and divisible modules are equivalent. In particular, let K be the fraction field of R , then since K is divisible as R -module, hence it's injective.

Given the inclusion $R \hookrightarrow K$, any index set I generates an inclusion $\bigoplus_{i \in I} R \hookrightarrow \bigoplus_{i \in I} K$ by coordinate wise inclusion. In particular, notice that $\bigoplus_{i \in I} K$ is also divisible (since for any nonzero $r \in R$, any $(x_i)_{i \in I} \in \bigoplus_{i \in I} K$ has $r \cdot (\frac{x_i}{r})_{i \in I} = (r \cdot \frac{x_i}{r})_{i \in I} = (x_i)_{i \in I}$). So, any quotient of $\bigoplus_{i \in I} K$ (as R -module) is also divisible, which is injective.

Now, for any R -module M , we know there exists some index set I , such that the free R -module $\bigoplus_{i \in I} R$ has a surjection onto M (for the most extreme case, choose $I := M$, and associate each element of M with a copy of R). Hence, one has $M \cong (\bigoplus_{i \in I} R)/L$ for some submodule $L \hookrightarrow \bigoplus_{i \in I} R$.

Then, consider the composition of inclusions $L \hookrightarrow \bigoplus_{i \in I} R \hookrightarrow \bigoplus_{i \in I} K$, this realizes L as a submodule of $\bigoplus_{i \in I} K$. Hence, one can consider the module theoretic quotient $(\bigoplus_{i \in I} K)/L$. Which, this generates an R -linear map $\varphi : \bigoplus_{i \in I} R \hookrightarrow \bigoplus_{i \in I} K \twoheadrightarrow (\bigoplus_{i \in I} K)/L$. Notice that one has an element $(a_i)_{i \in I} \in \ker(\varphi)$ iff $(a_i)_{i \in I} \in L$ (since when including into $\bigoplus_{i \in I} K$, it must lie in L), showing that $\ker(\varphi) = L$. As a result, it factors uniquely to an injective R -linear map $\bar{\varphi} : (\bigoplus_{i \in I} R)/L \hookrightarrow (\bigoplus_{i \in I} K)/L$:

$$\begin{array}{ccc} \bigoplus_{i \in I} R & \xrightarrow{\varphi} & (\bigoplus_{i \in I} K)/L \\ & \searrow \pi & \nearrow \bar{\varphi} \\ & (\bigoplus_{i \in I} R)/L & \end{array}$$

And, with $M \cong (\bigoplus_{i \in I} R)/L$, we have an inclusion of M into the injective R -module $(\bigoplus_{i \in I} K)/L$, finishing the desired claim.

12 HD (Type it up)

Problem 12

Let R be a commutative ring and let m be a given maximal ideal of R . Suppose M is an R -module such that $M_m = 0$. Prove or disprove that $M = 0$.

Solution: I guess we can take $R = \mathbb{Z}$, $m = 2\mathbb{Z}$, and $M = \mathbb{Z}/3\mathbb{Z}$. Then, M_m denotes all the elements of the form $\frac{\bar{a}_3}{d}$, where $d \notin 2\mathbb{Z}$ (i.e. we can divide by 3!) So in particular, one has the following:

$$3 \cdot (1 \cdot \bar{a}_3 - 1 \cdot \bar{0}_3) = 3 \cdot \bar{a}_3 = \bar{0}_3 \in M \quad (12.1)$$

This shows that $\frac{\bar{a}_3}{1} = \frac{0}{1}$ in M_m , showing $M_m = 0$, while $M \neq 0$.
(But if swapping to all maximal ideal m I think it's true).

13 HD (Type it up)

Problem 13

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be two ring homomorphisms such that S is R -flat and T is S -flat. Prove or disprove that T is R -flat.

Solution: Consider the following isomorphism as (R, S) -bimodules:

$$(R \otimes_R S) \otimes_S T \cong R \otimes_R (S \otimes_S T) \cong R \otimes_R T \cong T \quad (13.1)$$

As a result, if $I \hookrightarrow R$ is inclusion of ideal, one has the injection $I \otimes_R S \hookrightarrow R \otimes_R S \cong S$ as R -module (which is also an injective S -linear map, bc S is an (R, S) -bimodule). This is by R -flatness of S .

As a result, one has the following injection:

$$(I \otimes_R S) \otimes_S T \hookrightarrow (R \otimes_R S) \otimes_S T \cong S \otimes_S T \cong T \quad (13.2)$$

Which, using the bimodule isomorphism proven in problem 2, we have:

$$I \otimes_R T \cong I \otimes_R (S \otimes_S T) \hookrightarrow R \otimes_R (S \otimes_S T) \cong R \otimes_R T \quad (13.3)$$

Hence, T is also R -flat.

14 HD (Type it up)

Problem 14

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R -flat. Prove or disprove that φ can not be surjective.

Solution: Consider the isomorphism $\mathbb{Z}/30\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/5\mathbb{Z}$, I think this is also a $\mathbb{Z}/30\mathbb{Z}$ -mod. Then, take successive quotient we're good.

15 HD (Type it up)

Problem 15

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R -flat. Prove or disprove that ψ can not be surjective.

Solution: Look at 14 for counterexample, they apply to each other.

16 D

Problem 16

Let $R \xrightarrow{\varphi} S$ be ring homomorphism which is flat. Let M be an R -module such that $M \otimes_R S$ is torsion-free S -module. Is M torsion-free R -module?

Solution: We'll prove that M is not necessarily a torsion-free R -module, by constructing a counterexample.

Consider the ring $R := \mathbb{Z} \oplus \mathbb{Q}$ (with coordinate-wise addition and multiplication). Then, the projection onto the second coordinate $\pi_2 : R \twoheadrightarrow \mathbb{Q}$ defines $\mathbb{Q}$ as an R -algebra. Now, notice that as R -linear map, π_2 has a section, given as $j : \mathbb{Q} \rightarrow R$ as $j(q) = (0, q)$. Let's verify it has R -linear structure:

$$\forall q, r \in \mathbb{Q}, \quad j(q+r) = (0, q+r) = (0, q) + (0, r) = j(q) + j(r) \quad (16.1)$$

$$\forall (n, p) \in R, \quad j((n, p) \cdot q)j((\pi_2(n, p))q) = j(pq) = (0, pq) = (n, p) \cdot (0, q) \quad (16.2)$$

Hence, j is indeed an R -linear map. And, notice that any $q \in \mathbb{Q}$ satisfies $\pi_2 \circ j(q) = \pi_2(0, q) = q$, showing the projection π_2 splits.

This realizes $\mathbb{Q}$ as a direct summand of R when viewed as an R -module. Hence, $\mathbb{Q}$ is a direct summand of a free R -module, which is a projective R -module, in particular is flat.

Afterward, we'll denote π_2 as φ , and $\mathbb{Q}$ as S , so $\varphi : R \rightarrow S$ defines S as a flat R -module.

Now, consider $M := \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Q}$ as a ring (with coordinate-wise addition and multiplication again), and another projection map $\psi : R \twoheadrightarrow M$ by $\psi(n, p) := (\bar{n}, p)$ (where $\bar{n} \in \mathbb{Z}/2\mathbb{Z}$ is the quotient of n). This again realizes M as an R -module.

Notice that M as an R -module is not torsion-free: Consider the element $(2, 1) \in R$, and $(\bar{1}, 0) \in M$. The former element is not a zero-divisor in R , as if any $(n, p) \in R = \mathbb{Z} \oplus \mathbb{Q}$ satisfies $(2, 1) \cdot (n, p) = (2n, p) = (0, 0)$, one requires $p = 0 \in \mathbb{Q}$ and $2n = 0 \in \mathbb{Z}$, which also implies $n = 0 \in \mathbb{Z}$; on the other hand, $(\bar{1}, 0) \in M$ is a nonzero element. However, notice the action of $(2, 1) \in R$ on $(\bar{1}, 0) \in M$ is as follow:

$$(2, 1) \cdot (\bar{1}, 0) = (\psi(2, 1))(\bar{1}, 0) = (\bar{2}, 1)(\bar{1}, 0) = (\bar{0}, 1)(\bar{1}, 0) = (\bar{0}, 0) \quad (16.3)$$

So, the multiplication by $(2, 1)$ (an element that's not zero-divisor) is not injective, showing M has torsion elements as R -module.

Finally, we ought to prove that $M \otimes_R S$ is a torsion-free S -module. For this, let's recall the map $\psi : R \twoheadrightarrow M$ given by $\psi(n, p) = (\bar{n}, p)$. Hence, one has $(n, p) \in \ker(\varphi)$ iff $\bar{n} = 0 \in \mathbb{Z}/2\mathbb{Z}$ and $p = 0 \in \mathbb{Q}$, which is equivalent to $n \in 2\mathbb{Z}$ and $p = 0 \in \mathbb{Q}$. Therefore, $\ker(\varphi) = \{(2n, 0) \in R \mid n \in \mathbb{Z}\}$.

Now, consider the following exact sequence of R -modules:

$$0 \longrightarrow \ker(\varphi) \xrightarrow{\iota} R \xrightarrow{\psi} M \longrightarrow 0$$

where ι is the inclusion of ideals. By the flatness of S as an R -module proven before, we get the following exact sequence of R -modules:

$$0 \longrightarrow \ker(\varphi) \otimes_R S \xrightarrow{\iota \otimes \text{id}_S} R \otimes_R S \xrightarrow{\psi \otimes \text{id}_S} M \otimes_R S \longrightarrow 0$$

Here, we claim that $\ker(\varphi) \otimes_R S$ is 0: From above, its generators are of the form $(2n, 0) \otimes p$, where $(2n, 0) \in \ker(\varphi)$ (or $n \in \mathbb{Z}$) and $p \in S = \mathbb{Q}$. But, based on the way we define their R -module structure, we have the following:

$$\begin{aligned}
 (2n, 0) \otimes p &= ((1, 0)(2n, 0)) \otimes (1p) = (1, 0) \cdot_R ((2n, 0) \otimes (\varphi(0, 1)p)) \\
 &= (1, 0) \cdot_R ((2, 0) \otimes ((0, 1) \cdot_R p)) = (1, 0) \cdot_R ((0, 1) \cdot_R ((2n, 0) \otimes p)) \quad (16.4) \\
 &= ((1, 0)(0, 1)) \cdot_R ((2n, 0) \otimes p) = (0, 0) \cdot_R ((2n, 0) \otimes p) = 0
 \end{aligned}$$

(Note: To prevent confusion, here the R -action is denoted using $\cdot_R$, while the regular multiplication in each ring is without a dot).

Which, all generators of $\ker(\varphi) \otimes_R S$ is 0, showing that $\ker(\varphi) \otimes_R S = 0$. As a result, one has $R \otimes_R S \xrightarrow{\psi \otimes \text{id}_S} M \otimes_R S$ being an isomorphism by the exactness, and as R -module one has $R \otimes_R S \cong S$, so $M \otimes_R S \cong S$ as R -module.

Then, when realized as S -module, since $S = \mathbb{Q}$, then $M \otimes_R S \cong S$ is in fact a 1-dimensional S -vector space, which is torsion free as S -module.

Hence, the above example shows a ring homomorphism $R \xrightarrow{\varphi} S$ which is flat, M an R -module such that $M \otimes_R S$ is torsion-free as an S -module, yet M itself is not torsion-free as R -module. This provides a counterexample to the statement.